Demonstrating Retrocausal Wavefunction Collapse and Latent Momentum Uncertainty via Spatial Post-Selection of Single X-ray Photons

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Abstract

This study presents a macroscopic, hardware-based framework for testing the Two-State Vector Formalism (TSVF) of quantum mechanics through the delayed spatial post-selection of 1-Ångström X-ray photons. By employing a beryllium double slit to create an initial spatial superposition, followed by an asymmetric silicon Bragg crystal to induce a transverse phase shift, the setup divides the coherent wavefront using a multi-layer 50/50 X-ray beam splitter. The experimental post-selection choice is executed on the reflected branch via a highly restrictive 1.5-μm spatial shift aperture. Computational models combining forward time evolution with backward-propagating constraints demonstrate that measuring the particle at this precise spatial window retroactively dictates its path through the initial double slit, effectively erasing the superposition. Furthermore, Monte Carlo simulations of far-field single-photon detection using a Sodium Iodide (NaI) scintillation detector reveal that the extreme position constraint at the aperture induces a latent, uncertainty-driven momentum spread. The statistical accumulation of these discrete far-field clicks maps precisely to a continuous diffraction pattern, isolating the retrocausal collapse to the single-particle regime and providing observable evidence of backward-in-time constraints on quantum histories.

Materials and Basic X-ray Detector Equipment Construction

1. Primary Optical Path Materials

• **X-ray Source:** Laboratory-grade X-ray tube with a heavily attenuated output to ensure single-photon emission at approximately 1 Ångström (12.4 keV energy).

• **Beryllium Double Slit:** Precision-machined beryllium foil with slits separated by 20 μm, chosen for its high X-ray transparency.

• **Asymmetric Bragg Crystal:** Polished silicon crystal cut asymmetrically to the lattice planes to act as a spatial compressor and induce the necessary transverse phase shift.

• **Multi-layer X-ray Beam Splitter:** Housed in a brushed steel "strongbox" to act as the experimental chance box, yielding a 50/50 reflection/transmission ratio.

• **Spatial Shift Aperture:** Micrometer-adjustable metallic slit mechanism, calibrated precisely to a 1.5-μm window to force extreme spatial post-selection.

2. Detector Equipment and Construction

To capture the individual far-field photon clicks and isolate the ~12.4 keV energy signals from background noise, a custom scintillation detector must be constructed:

• **Scintillator Selection:** A Thallium-doped Sodium Iodide (NaI(Tl)) crystal is selected for its high light yield and excellent stopping power for 12 keV X-rays.

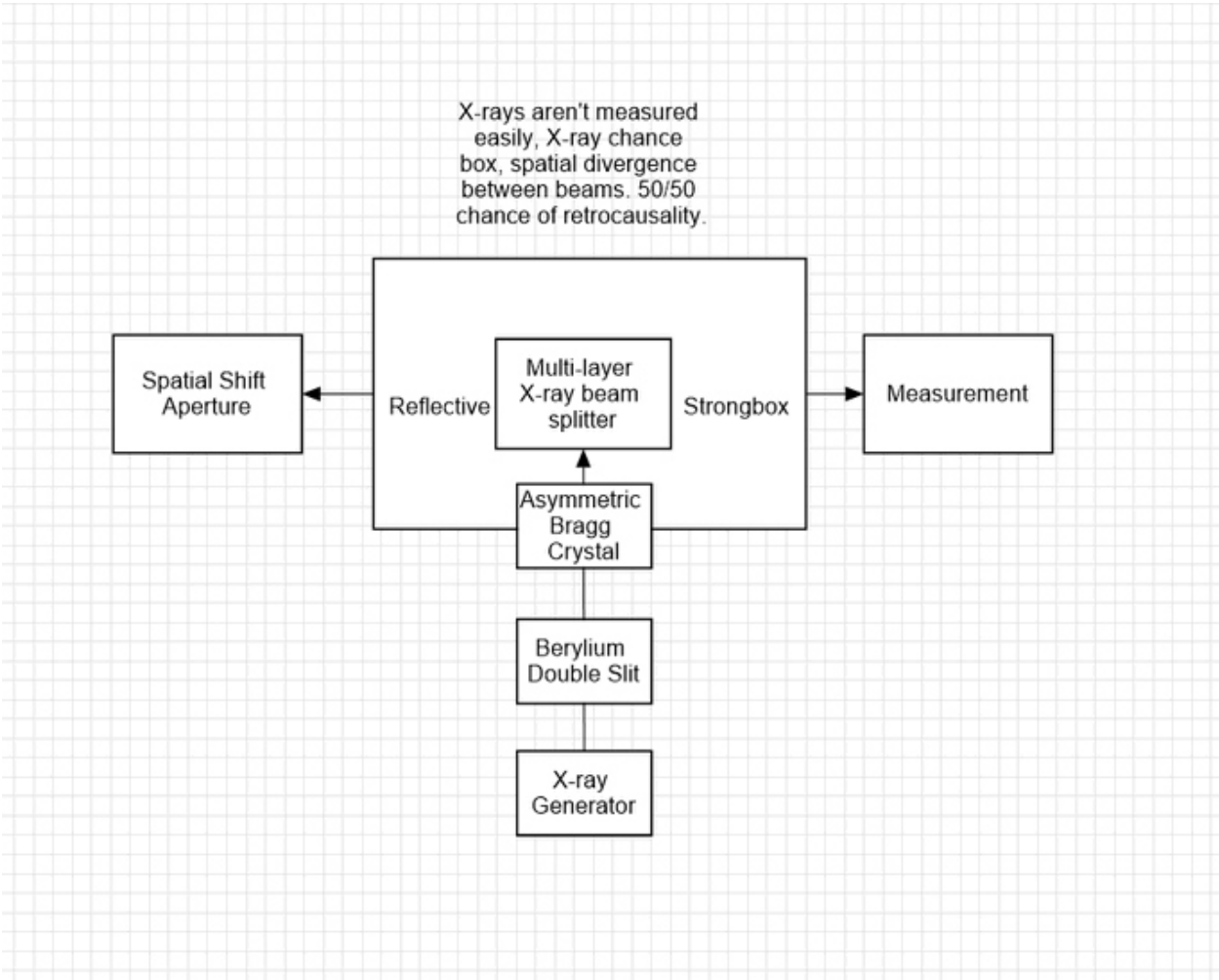
• **Optical Coupling:** The NaI(Tl) crystal is mated to the input window of a highly sensitive Photomultiplier Tube (PMT) or Silicon Photomultiplier (SiPM) array using a specialized, UV-transparent optical grease to maximize photon transfer.

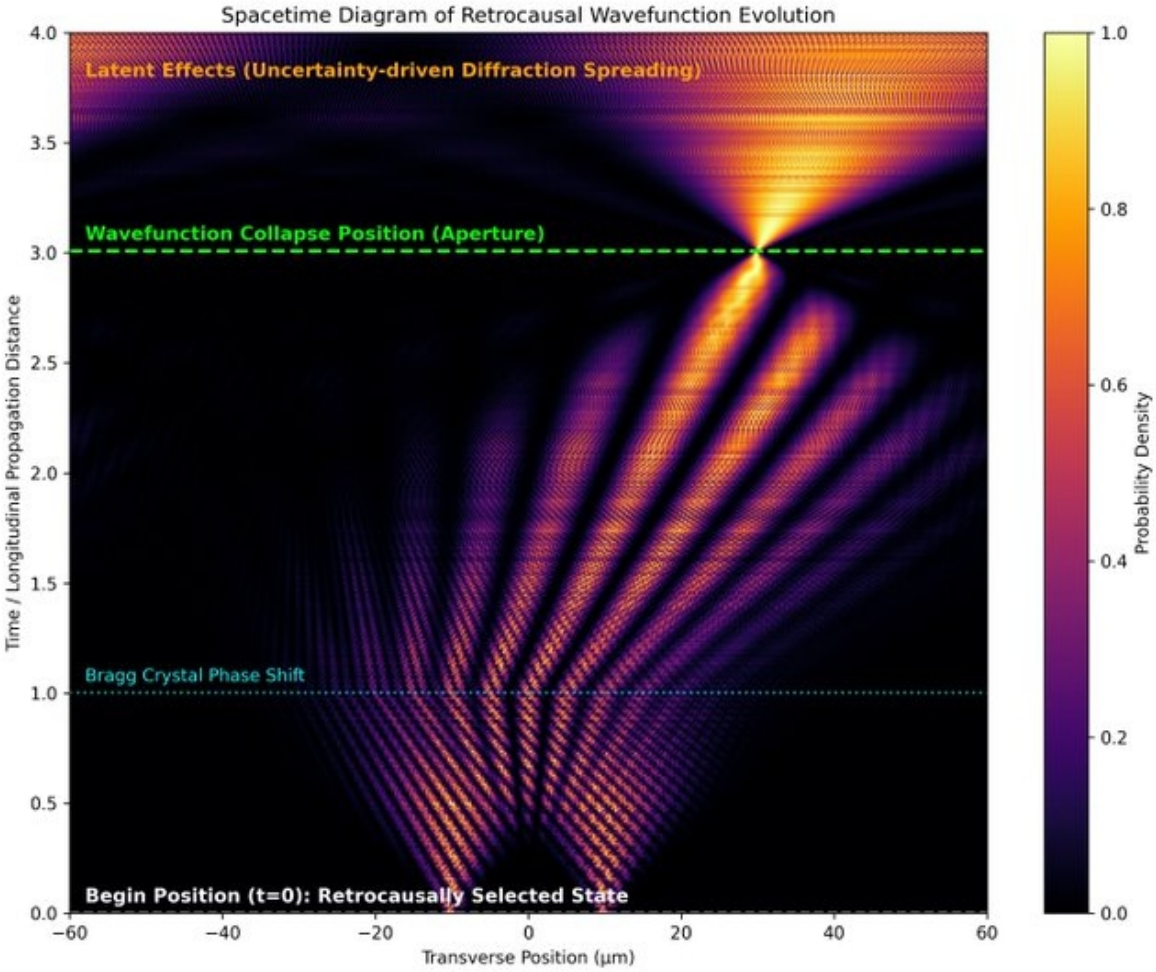
• **Light-Tight Housing:** Because the PMT can be triggered by a single photon of ambient light, the entire scintillator-PMT assembly is wrapped in white Teflon reflector tape (to direct scattered scintillation photons into the PMT) and sealed inside a black-anodized, light-tight aluminum cylindrical housing.

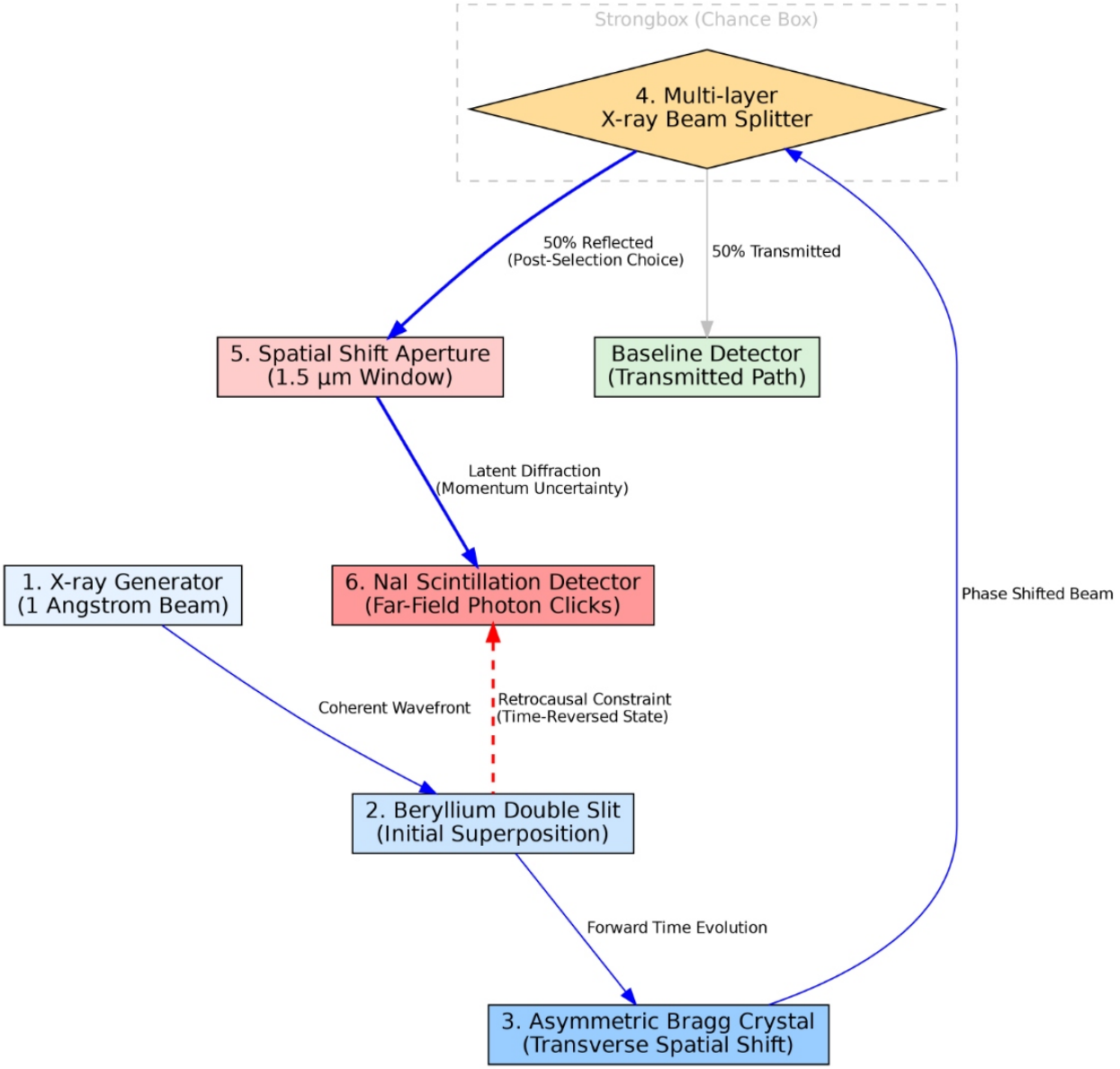
• **Electronics Integration:** The PMT is connected to a high-voltage power supply. The output signal is routed via thick coaxial cables through a preamplifier, and then into a Multi-Channel Analyzer (MCA) or digital oscilloscope for precise energy thresholding.

• **Alignment:** The detector housing is mounted on a motorized, precision linear translation stage situated 50 cm away from the spatial shift aperture. This allows the system to slowly scan across the transverse axis, mapping the latent diffraction pattern click-by-click.

Experimental Diagrams


X-rays aren't measured easily, X-ray chance box, spatial divergence between beams. 50/50 chance of retrocausality.
Spatial Shift Aperture
Reflective
Multi-layer X-ray beam splitter
Strongbox
Measurement
Asymmetric Bragg Crystal
Berylium Double Slit
X-ray Generator


Spacetime Diagram of Retrocausal Wavefunction Evolution
Latent Effects (Uncertainty-driven Diffraction Spreading)
Wavefunction Collapse Position (Aperture)
Bragg Crystal Phase Shift
Begin Position (t=0): Retrocausally Selected State
Time / Longitudinal Propagation Distance
Transverse Position (μm)
Probability Density
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−60
−40
−20
0
20
40
60
1.0
0.8
0.6
0.4
0.2
0.0


Strongbox (Chance Box)
4. Multi-layer
X-ray Beam Splitter
50% Reflected
(Post-Selection Choice)
50% Transmitted
5. Spatial Shift Aperture
(1.5 µm Window)
Baseline Detector
(Transmitted Path)
Latent Diffraction
(Momentum Uncertainty)
1. X-ray Generator
(1 Angstrom Beam)
6. NaI Scintillation Detector
(Far-Field Photon Clicks)
Phase Shifted Beam
Coherent Wavefront
Retrocausal Constraint
(Time-Reversed State)
2. Beryllium Double Slit
(Initial Superposition)
Forward Time Evolution
3. Asymmetric Bragg Crystal
(Transverse Spatial Shift)